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PI Name:	Mitchell, Cary Ph.D.		
Project Title:	(NSCORT) for Advanced Life Support at Purdue/Howard/AAMU - Food Safety and Food Processing		
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PI Email:	cmitchel@purdue.edu	**Fax:**	FY 765-494-0391
PI Organization Type:	UNIVERSITY	**Phone:**	765-494-1347
Organization Name:	Purdue University		
PI Address 1:	Department of Horticulture and Landscape Architecture		
PI Address 2:	625 Agriculture Mall Dr		
PI Web Page:	http://www.alsnscort.org		
City:	West Lafayette	**State:**	IN
Zip Code:	47907-2010	**Congressional District:**	4
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COI Name (Institution):	Applegate, Bruce (Purdue University) Mauer, Lisa (Purdue University) Williams, Leonard (Alabama A&M University)		
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Task Description:	In an ALS, the food subsystem is an integral part of the mission. Purdue and Alabama A&M Universities are continuing to address key components of food processing, food safety, energy requirements, and the effect of long term exposure of food to radiation.. Currently there are three major efforts addressing these food related issues for a potential NASA mission to Mars or establishment of a lunar base. It is also important to consider that the research, although space based, has terrestrial benefits as well. To address food safety, research is ongoing in both specific detection of pathogens and prevention of contamination in food processing. By continuing to identify potential areas of contamination and control strategies using an integrated HACCP (hazard analysis and critical control points) approach key parameters are being identified from harvest through processing and consumption. The HACCP approach has also been recently expanded to evaluate the ALS system as a whole for identification of key parameters and possible points of contamination from the various waste processing regimes, hydroponics, and tilapia waste water. Research has also shown proof in principle experiments of a phage based immunoassay allowing signal amplification by exploiting phage amplification in a bacterial host. This approach will allow a simple test for pathogen detection which can be implemented with minimal equipment and training and has potential for impacting food safety and biosecurity in a terrestrial application. Research on effects of radiation on foods has also begun to understand impacts of prolonged exposure on quality of food including the nutritional and sensory aspects. Bruce Applegate Focus Area Group Leader – Food Safety and Processing The goal of this research is to harness the power of bacterial phage display to develop a biological amplifier for the detection of small numbers of pathogenic organisms in potable water and foods. The research will generate and purify phages that are designed to selectively detect water and foodborne pathogens. Using specially designed markers that are inserted into the DNA of the phage, the expression of the phage can be tuned to reflect the presence (or absence) of the target cell as well as give a measure of concentration and type of cells in a given sample. Proteins displayed on the outside of the phages can be used as “handles” to distinguish one type of phage from another, much like a license plate will identify one car from another, even if the cars are alike in every way except for the owner. Since the protein is displayed on the outer surface of the phage and since it reflects a special characteristic of the DNA contained inside the phage particle, this is referred to as phage display. The proposed project combines the knowledge and experiences of fundamental molecular biology of phages and applies this knowledge to existing technologies in both the molecular construction and detection of the modified bacteriophages. The assay will provide an alternative to current culturing methodology and nucleic acid amplification technology which are time, labor, and equipment intensive. The fundamental hypothesis of this research is: Bacteriophage can be genetically modified using recombinant DNA technology to produce an antigenic peptide which is only expressed in progeny phage after infection of a specific viable pathogenic organism. The hypothesis will be tested by addressing the following objectives: 1. Genetically modify and propagate modified bacteriophages for the detection of viable pathogens. 2. Recover and purify modified bacteriophages using affinity chromatography. 3. Develop an assay using the modified bacteriophages to detect pathogenic organisms in potable water and food using an ELISA format. 4. Integrate the developed bacteriophage assay into novel food packaging materials. For Food Processing and Packaging: Food must be safe, nutritious, and acceptable throughout a long duration mission. In addition to a pre-packaged food supply, the ALS NSCORT proposes to utilize crops to provide food and oxygen for astronauts during long-term space missions. Research is required to better understand the ability to convert edible biomass into safe, nutritious, and acceptable food products in a closed system with many restrictions (mass, volume, power, crew time, etc.). As presented in the original NSCORT proposal, the objectives for this project are to: 1. Utilize biomass produced by Purdue and AAMU researchers to produce food products and ingredients and generate solid and liquid wastes from food processing and packaging operations. All mass, energy, and ESM data will be provided to the systems analysts. 2. Define, model, then minimize ESM for unit operations used in food production 3. Measure food acceptability, proximate or key nutrient composition, and select functional properties of products and ingredients produced in objectives 1 and 2. 4. Define, model, then minimize ESM and residual waste for packaging systems used for foods. 5. Develop an integrated package-safety indicator system for extended shelf-life foods incorporating biosensor technology with Dr. Applegate in years 3-5. Based on discussions and recommendations from the external advisory committee review in November, 2003, the focus of projects will shift from an applied emphasis on ESM of the food system to a basic research emphasis on quality of foods and effects of radiation on foods. For Optimal Food Safety: The primary goal of this project is to develop a HACCP system which can be used for the validation and testing of food products for ALS mission and determine the critical points in processing and production of packaged products provided from plants produced by Purdue and AAMU researchers, and develop a HACCP and food safety system to monitor and validate the effectiveness of reducing any hazards in these packaged products. In turn, the associated objectives would be as follows: 1. Develop a sampling plan in coordination with Purdue researchers (Mauer and Applegate) at each critical point in processing and packaging products. 2. Define the monitoring, critical limits and corrective actions of pathogenic organisms (L. monocytogenes, Campylobacter jejuni, Salmonella, E. coli O157:H7) isolated from food systems. 3. Develop integrated HACCP plans for potable water and foods to validate the effectiveness of objective 1 and 2 4. Develop an integrated HACCP system into novel food safety and HACCP plans for non-food safety and microbiology personnel.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Further development of nano-scale detection of pathogens in packaged foods Development of novel food products and technologies for vegetarian and health-food markets
	For Pathogen Detection in Potable Water and Food: Cumulative Research Progress to Date Modified bacteriophages for the detection of E. coli O157:H7 and Salmonella spp. are currently being constructed for the detection platform. Details of the progress are described below: Salmonella spp bacteriophage: To facilitate the construction of a modified P22 bacteriophage for Salmonella spp. it was necessary to construct a recombination vector for insertion of a modified tail spike protein. The vector was modified from a previously constructed vector pTP369 (Casjens et al). Plasmid pTP369 contains a 2558 bp region of the P22 genome corresponding to the region containing gene 23 (antitermination protein), gene 13 (lysis protein), gene 19 (lysozyme) gene 15 (lysis control), orf 201 (unknown protein) and orf 80 (unknown protein). The recombination vector was constructed by removing approximately 1 kb of the P22 genome in the region of orf 201 and orf 80 to allow the

Task Progress:

insertion of DNA for recombination. A multicloning site and a TA cloning site were inserted to facilitate rapid insertion of modified DNA to construct the appropriate epitope. Primers were utilized to amplify the tailspike protein from P22 with the appropriate His modifications. We are currently adding the appropriate promoter configurations to allow repressed expression of the His modified tailspike protein in the preparative host strain. The preparative host strain repressor gene cassette was constructed for insertion into the preparative host strain genome. Initial work was begun on inserting the modified tailspike gene in the previously constructed P22 recombination vector. Appropriate promoter and terminator configurations were added to the modified tailspike protein to allow repression of expression of the His modified tailspike protein in the preparative host strain. Preparative host strain containing the lacI repressor gene cassette was constructed. The strain is currently being evaluated for sufficient repression activity using a lac-lux fusion. The resultant recombination construct was utilized to facilitate a development of a detection system utilizing bacteriophage pathogen-specificity and a bacterial bioluminescent bioreporter utilizing the quorum sensing molecule from Vibrio fischeri, N-(3-oxohexanoyl)-homoserine lactone (3-oxo-C6-HSL) for the detection of Salmonella sp. The assay exploits phage host specificity coupled with a modified phage genome to reprogram the host cell to express acyl-homoserine lactone synthetase (luxI) leading subsequent production of 3-oxo-C6-HSL. The 3-oxo-C6-HSL molecules diffuse out of the target cell resulting in induction of bioluminescence from a population of 3-oxo-C6-HSL inducible bioreporters (RO/lux). Assays utilizing the luxI/ROlux based phage system for the detection of Salmonella enteriditis showed detection limits of 1000 bacteria in approximately 6 hours using a portable luminometer.

E. coli O157:H7 bacteriophage: Phage based detection of E. coli O157:H7 will be accomplished using bacteriophage ?V10. Bacteriophage ?V10 was originally isolated by R. Khakhria and has been shown to specically infect many strains of E.coli O157:H7. It has a genome of approximately 42 kb and classified as a temperate phage (can form lysogens). The phage was obtained from Dr. Rafiq Ahmed at the National Laboratory for Enteric Pathogens in Winnipeg Canada. To determine the specificity of ?V10, a previously characterized library from pathogenic E. coli outbreaks responsible for disease was screened for ?V10 susceptability. Environmental isolates were also screened to provide evaluation of false positives. The assay consists of a simple plaque assay using a previously prepared phage solution with a titer of approximately 2 x 10 -3 plaque forming units per mL. Using this assay, 187 strains of E. coli were tested. Of these samples, 106 were known to be isolates of E. coli O157:H7 and 81 were not. Results showed 93 positives and 81 negatives. The data correlated with the previous identifications. However 13 of the previously identified positives did not form plaques and were investigated further and found to be contaminants. No false positives were detected. Total DNA was isolated from the phage and a genomic library for sequencing the entire phage genome was generated. The library was sequenced providing data which will allow a more deliberate construction of the recombinant phage avoiding the use of transposon mutagenesis. (Note: An anomaly in the previously reported sequencing data suggests a unique DNA packaging requirement for phi V10 which is being investigated. It appears the phage may have a tripartite genome. If this is the case it would allow large insertions of foreign DNA.) Preliminary sequencing generated two contigs of 22 kb and 15 kb. Sequence analysis showed high homology to the Enterobacterial phage epsilon 15 and numerous E. coli O157:H7 bacteriophage. The final sequencing is currently being performed to resolve ambiguities in the genome along with completing the annotation of the genome.

A similar recombination system for insertion of foreign DNA into the phage genomes as described for the P22 above was developed for the modification of the E. coli O157:H7 bacteriophage FV10. This was accomplished by isolating a FV10 lysogen of E. coli O157:H7. The lysogen was further characterized and shown to be inducible for the lytic phenotype. The strain was subsequently transformed with pKD46 which contains an arabinose inducible recombinase. This system allows the electroporation of linear DNA containing foreign genes flanked by DNA sequences from regions of the phage genome. This allows insertion of reporter genes into specific regions of the phage genome. Initial work has identified regions for modification, allowing insertion of reporter genes which do not affect the phage life cycle. Although designated for reporter gene insertion we are also utilizing the system for analysis of the unknown open reading frames found in the genome of bacteriophage FV10. Key parameters were determined for optimization of the recombination system for reporter gene insertion and gene inactivation. Primers were designed to begin the systematic evaluation of the unknown open reading frames. Amplicons were constructed for the systematic disruption of identified open reading frames in phi V10. We are currently beginning the recombination and lysogen rescue experiments.

Listria monocytogenes bacteriophage: Preciaus Heard (Summer Fellowship Student) began initial screening of a Listeria monocytogenes library to isolate a lysogenic bacteriophage. Preliminary results suggest she was successful and we are currently further evaluating the isolated bacteriophage and continuing to screen the library.

For Novel Food Processing and Packaging: Cumulative Research Progress to Date

• Equivalent System Mass (ESM) Estimates for Commercially Available, Small-Scale Food Processing Equipment (work completed and presented at the ICES annual meeting) One of the challenges NASA faces today is developing an Advanced Life Support (ALS) system that will enable long duration space missions beyond low earth orbit (LEO). This ALS system must include a food processing subsystem capable of producing a variety of nutritious, acceptable, and safe edible ingredients and food products from pre-packaged and re-supply foods as well as salad crops grown on the transit vehicle or other crops grown on planetary surfaces. However, designing, building, developing, and maintaining such a subsystem is bound

to many constraints and restrictions. The limited power supply, storage locations, variety of crops, crew time, need to minimize waste, and other ESM parameters influence the selection of processing equipment and techniques. Several researchers have calculated ESM of select types of food processing equipment to compare ESM for individual food types; however, a complete survey of ESM parameters for currently available food processing unit operations has not been completed. In order to direct NASA’s research and technology efforts related to the food subsystem, the technologies available on Earth for food processing, preservation, and packaging must be identified and the viability of these technologies must be assessed. Minimizing mass, volume, and energy consumption are important factors to be considered when locating available food processing equipment and evaluating feasibility for use in an ALS system. Once the ESM has been estimated for available equipment, modifications can be suggested to improve efficiency and reduce ESM. The objective of this study was to compile ESM-parameter information (mass, volume, and power) for currently available, small-scale food processing equipment and to provide average, high, and low ESM values for each class of equipment (hand-held and bench-top mixers, etc.) that performs the following unit operations: mixing, size reduction, heat transfer (heating and cooling), and extraction (water, oil, and juice). In this study, each piece of equipment was assumed to perform a single task, the power required for cooling was set equivalent to the power needed to operate the equipment, and the crew-time was not considered in the preliminary ESM estimates. ESM (non-crewtime) estimates were also provided for different mission scenarios. A database of this information has now been created and is available for further development and updates. The data and ESM estimates presented will provide a starting point for

the minimization of the ESM of food processing equipment for a long term mission. This work was done in collaboration with Dr. Michele Perchonok at JSC.

• Equivalent System Mass of Producing Yeast and Flat Breads From Wheat Berries, A Comparison Of Mill Type (work completed and presented at the ICES annual meeting)

This work built on a study presented at the 2003 ICES meeting entitled “Comparison of Equivalent System Mass (ESM) of Yeast and Flat Bread Systems”. Wheat is a candidate crop for the Advanced Life Support (ALS) system, and cereal grains and their products will be included on long-term space missions beyond low earth orbit. While the exact supply scenario has yet to be determined, some type of post-processing of these grains must occur if they are shipped as bulk ingredients or grown on site for use in foods. Understanding the requirements for processing grains in space is essential for incorporating the process into the ALS food system. The ESM metric developed by NASA describes and compares individual system impact on a closed system in terms of a single parameter, mass. The objective of this study was to compare the impact of grain mill type on the ESM of producing yeast and flat breads. Hard red spring wheat berries were ground using a Brabender Quadrumat Jr. or the Kitchen-Aid grain mill attachment (both are proposed post-harvest technologies for the ALS system) to produce white and whole wheat flour, respectively. Yeast bread was made using three methods (hand+oven, bread machine, mixer with dough hook attachment + oven). Flat bread was made using four methods (hand+oven, hand+griddle, mixer+oven, mixer+griddle). Data on all inputs (active time, passive time, mass and volume of ingredients and equipment, power) were measured and used to calculate ESM. Assumptions were based on data in NASA documents. Data were analyzed using PC-SAS with significance at $P < 0.05$. Grain mill type significantly ($P < 0.05$) influenced the ESM of making both bread types; and the Brabender Quadrumat Jr. contributed significantly ($P < 0.05$) more mass than the Kitchen-Aid grain mill to the ESM for producing both types of bread. Additionally, the approach used for measuring ESM for a food production scenario can be used for evaluating ESM of producing any food and can be a useful template for future investigations. This work was done in collaboration with Dr. Michele Perchonok at JSC.

• Characterization of wheat cultivars intended for growth during long-term space missions and comparison to select common terrestrial cultivars (preliminary study presented at the IFT annual meeting)

Cereal grains and their products will be included in long-term space missions beyond low earth orbit. Wheat is a candidate crop for the Advanced Life Support system and will likely be grown with other crops to provide food, oxygen, and water purification during extended planetary research missions. Apogee and Perigee are hard red spring wheat cultivars with dwarf and super-dwarf heights, respectively. These wheat varieties were developed at Utah State University for growth in space. Unique characteristics of these cultivars include: short height at full maturity, high edible bio-mass production per unit area, and elevated protein levels. The chemical and physical characteristics of these wheat cultivars important for functionality in foods have not been thoroughly investigated. The objective of this study was to characterize Apogee and Perigee wheat cultivars and compare them to common terrestrial wheat varieties using chemical and physical assays. Apogee and Perigee were grown hydroponically and in the field at Utah State University. Yecora Rojo, Parshall, and Yavaros 79 were purchased from commercial seed suppliers. Yecora Rojo and Parshall are hard red spring wheats, while Yarvaros 79 is a durum wheat. Chemical assays conducted on all wheat varieties were: proximate analysis, non-protein nitrogen, nitrate content, free-amino acid content, insoluble polymeric protein content, and total antioxidant capacity. The physical assay conducted on all wheat varieties was the Farinograph. Wheats were compared by wheat type, growth media (field vs. hydroponic), and flour type (whole-wheat vs white) for these assays. Wheat grown hydroponically exhibited higher protein levels than field grown wheat. Non-protein nitrogen was not accumulated as urea or ammonia. Proximate composition, nitrate levels, insoluble polymeric protein content, and total antioxidant capacity varied between varieties and growth conditions. Based on preliminary results, Perigee would be the best wheat variety for growth in space because of its short stature and high protein levels; however replication of this study is needed for statistical evaluation of the data. This work was done in collaboration with Dr. Bruce Bugbee at Utah State University and Dr. Michele Perchonok at JSC.

For Optimal Food Safety: Research has begun on objective one and two of the above section, development of sampling plans, validation and monitoring plans for pathogenic bacteria. Lettuce, carrot and tomatoes are currently being used to determine the efficacy of commonly used sanitizers and antimicrobial for reducing or eliminating spoilage and pathogenic bacteria at both pre and post harvest and to increase the shelf-life of each salad crop.

Efficacy of sanitizing agents on reduction of Salmonella on the surface of tomatoes: To determine the role of sanitizing agents on reduction of an high inoculum of Salmonella spp. on surfaces of tomatoes, we decided to used several sanitizers that are generally recognized as safe (GRAS). Four sanitizers, peracetic acid, hydrogen peroxide, chlorine and Proscan®, a commercially available product were all chosen based on their potency, FDA approval for use and its ability to leave behind limited amounts of chemical residues. Two stains of Salmonella were used in this study to determine the ability of biofilm formation, increased ripening or spoilage and their storage at room temperature on short period of time (up to 6 days of storage). All strains were provided by the AAMU Food Microbiology Laboratory and each were cultured separately in tryptic soy broth (TSB) at 37 C for 24 h with agitation, washed, and resuspended into 0.1% peptone water. In this project, we decided to optical density of 0.5 at 640 nm (representing approximately 109 CFU/ml). Six dipping trials on tomatoes were conducted, each with single test pH within the range of 4 to 9. In each trail, 50 tomatoes were divided into groups. Each group consisted of 6 samples, 2 as controls and 4 sanitizer treatments. Group 1 and 6 was used without sanitizer treatment to determine the visual color changes of tomatoes after treatment. Groups 2, 3, 4 and 5 were inoculated with Salmonella spp. to determine the reduction of individual disinfectants on survival of Salmonella due to treatments. Samples were dried under laminar flow hood to dry surface for 1 h. For treatment of tomatoes, samples were dipped into a 2 L sterile sample bag containing 800 ml of each sanitizer for 3 min at 25° C. Treatment temperatures were achieved by storing samples in temperature controlled incubator.

Microbial enumeration: Each control and treatment were suspended in 225 ml of 0.1% buffered peptone water in a stomaching bag, and gently agitated for 120 s. Stomaching waters of each control were collected, diluted and plated onto 100 mm plates, while the stomaching waters of each treatment were plated directly onto tryptic soy agar and incubated overnight. For better recovery of injured cells, colonies were counted again after incubation after incubation for another 24 h. Evaluation of the color of tomatoes: the color of the tomatoes on the outer surfaces and ripening (softening) were measured by using a colorimeter and by visual inspection.

Data can be summarized as follows: the numbers of Salmonella bacteria recovered from tomato samples were dramatically reduced by 0.91 CFU/ tomato log reductions from the initial population when observed at 2 days of storage at room temperature

	However, at 4 d of storage, the survival of Salmonella on the surface of tomatoes increased with an increase in the storage time. Suggesting that Salmonella acquired the ability to adjust to the changes in the pH of the sanitizers, recover and eventually start attaching to the surfaces of the tomatoes No obvious color changes of tomatoes were visually observed after sanitizer treatment
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Articles in Other Journals or Periodicals	Gandolph, J., M.G. El-Abiad, M. Perchonok, L.J. Mauer. " Equivalent System Mass (ESM) estimates for commercially available, small-scale food processing equipment. " Proceedings of the SAE International Meeting, Colorado Springs, CO, July 19-22.This paper was also selected for publication in SAE Transactions)., Paper # 2004-01-2526 , Jan-2004
Articles in Other Journals or Periodicals	Weiss, I., M. Perchonok, K.D. Hayes, and L.J. Mauer. "ESM of producing yeast and flat breads from wheat grains: a comparison of grain mill type." Paper # 2004-01-2525. , Jul-2004
Papers from Meeting Proceedings	Weiss I, Ozen BJ, Perchonok M, Hayes KD, Mauer LJ. "Comparison of Equivalent System Mass (ESM) of yeast and flat bread systems." 33rd International Conference on Environmental Systems, Vancouver, BC, Canada July 7-10, 2003. SAE Paper 2003-02-2618, Warrendale, PA: SAE International, 2003. , Jul-2003
Patents	Provisional Provisional patent filed February 2004 Feb-2004 Bruce Applegate and Michael Ladisch. "Phage Detection of Pathogens (PDP System) "
Presentation	I. Weiss, K.D. Hayes, M. Perchonok, and L.J. Mauer. "Comparison of equivalent system mass (ESM) and food metric value (FMV) of yeast and flat bread systems. " Institute of Food Technologists' Annual Meeting and Food Expo. July 12-16, Chicago, IL. (poster) Jul-2003
Presentation	M. Shroyer, U. Minocha, I. Weiss, L. Williams, L.J. Mauer, B.M. Applegate. "NSCORT food system and safety design for a NASA mission to Mars" Habitation 2004. Jan. 4-7, Orlando, FL. (poster) Jan-2004
Presentation	I. Weiss, K.D. Hayes, B. Bugbee, M. Perchonok, L.J. Mauer. "Characterization of wheat cultivars intended for growth during long-term space missions and comparison to select common terrestrial cultivars. " Institute of Food Technologists' Annual Meeting and Food Expo. July, Las Vegas, NV. (poster) Jul-2004
Presentation	L.J. Mauer "Mars, Food, and Agriculture" ". Council for Agricultural Science and Technology VIP Day, Purdue University, West Lafayette, IN, January 17. (invited talk, keynote speaker) Jan-2003
Presentation	L.J. Mauer. "Food for space travel" Wabash Area Lifetime Learning Association, Inc. Continuing Education Program, West Lafayette, IN. October 20. Oct-2004
Presentation	Shroyer, M., U. Minocha, M. Ladisch, and B. Applegate. "Bioamplification using phage display for the detection of pathogens. " American Society for Microbiology Conference on the New Phage Biology. Key Biscayne, FL. Sep-2004
Presentation	Minocha, U., R. Jennings, A. Bhunia, and B. Applegate. "Genome sequence analysis and evaluation of strain specificity of E. coli O157:H7 bacteriophage phi V10. " American Society for Microbiology, New Orleans, LA. Jun-2004
Presentation	Shroyer, M., U. Minocha, N. Bright, L. Perry, and B. Applegate ". Development of a recombination system for rapid construction of E. coli O157:H7 reporter bacteriophage" American Society for Microbiology, New Orleans, LA. Jun-2004
Presentation	Minocha, U., N. Bright, L. Perry, and B. Applegate "Detection of the foodborne pathogen Escherchia coli O157:H7 using an AINS recombinant phiV10 bacteriophage based bioluminescent reporter system." American Society for Microbiology Biodefense Research Meeting. Baltimore, MD. Sep-2004